KUKU FM | AI ENGINEERING CHALLENGE | PROBLEM 1

# Autonomous Promo Finder

From 50 screenplay episodes to a ranked, verified, ready-to-post shortlist of 15 promo cuts.

SUBMITTED BY
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Solution Document | November 2026

| 50 / 3,727 | 15 | 100% | $0.0934 |
|---|---|---|---|
| **Episodes / Elements** | **Shortlisted Promos** | **Boundary Verified** | **Total Compute** |
| 100% parsed AST screenplay elements | Covering 14 unique episodes, 20-60s bounds | Zero mid-sentence cuts, strict timestamp math | <$0.002 / episode; >95% human labor reduction |

## 01 END-TO-END PIPELINE ARCHITECTURE

LLM proposes. Deterministic code disposes.

1 AST PARSER → 2 MOMENT MINER → 3 MULTI-SCORER → 4 ADVERSARIAL AUDIT → 5 DIVERSITY SHORTLIST

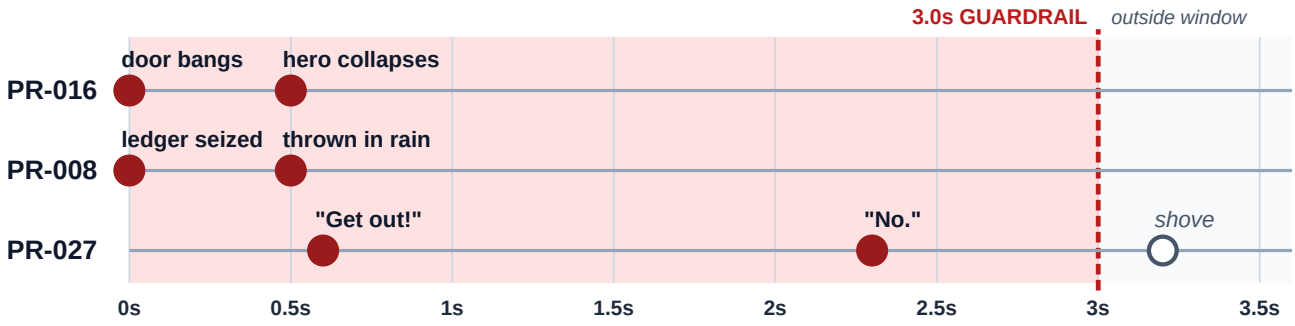
| # | STAGE | WHAT IT DOES |
|---|---|---|
| 1 | **Deterministic AST Parser** | Normalizes UTF-8 scripts into scenes, actions and timed dialogue. Assigns **immutable element IDs**. |
| 2 | **Moment Mining & Stitcher** | LLM finds high-tension anchors. Code stitches continuous or **non-contiguous cuts** (e.g. 2-segment pacing) with no invented timecodes. |
| 3 | **Multi-Criteria Scorer** | Hook **30%**, Cliffhanger **25%**, Standalone Coherence **20%**, Emotion **15%**, Duration Fitness **10%**. |
| 4 | **Adversarial Audit & Policy** | Validates hook inside **t <= 3.0s**, detects premature suspense resolution, checks pronouns, scans platform-risk heuristics. |
| 5 | **Diversity & Anti-Fatigue** | Picks top 15 with **max 2 promos / episode** and varied angles: Revenge, Survival, Identity. |

## 02 TOP PROMO STRATEGY MATRIX

Ranked by composite score, posting order included

| RANK | CUT | SOURCE & TIMING | SCORE | ANGLE | HOOK (t <= 3s) | CLIFFHANGER | POSTING RATIONALE |
|---|---|---|---|---|---|---|---|
| **#1** | **PR-016** | Ep10<br>00:02.10-00:38.90<br>**36.8s** | **8.42** | **Survival / Mystery** | Door bangs open **(+0.0s)**; blood-soaked hero collapses in doorway **(+0.5s)**. Immediate sonic and visual scroll-stopper. | Juliana gasps *"Dear God"* as wounds are exposed; cut **before treatment or explanation**. | **Post 1st.** Maximum scroll-stopping shock with a simple 2-person cold-start dynamic. |
| **#2** | **PR-008** | Ep5<br>00:54.00-01:14.50<br>+ 01:31.00-01:40.80<br>**30.3s** (2 segments) | **8.35** | **Betrayal / Revenge** | Heirloom ledger wrenched away **(+0.0s)** and thrown into rain **(+0.5s)**. Visceral cruelty. | *"Choose your next words carefully"* warning; cut **before physical escalation**. | **Post 2nd.** Highest active-confrontation cliffhanger, stitched across 2 segments. |
| **#3** | **PR-027** | Ep15<br>00:44.20-01:26.00<br>**41.8s** | **8.18** | **Protector / Action** | Thug demands *"Get out of the way!"* **(+0.6s)**; Juliana defies *"No."* **(+2.3s)**. Shove follows at +3.2s. | Thug challenges *"You want to fight us, veteran?"* Cut on the breath **before the brawl**. | **Post 3rd.** Classic protector arrival with strong narrative conflict. |

**Hook beats on the clock** (filled = inside the 3.0s guardrail)

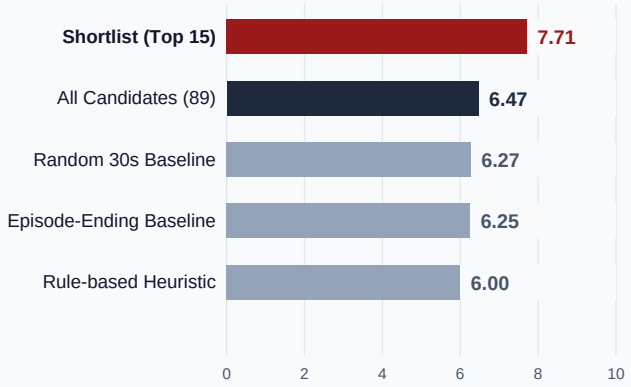



**Why this order?** Lead with the fastest-reading shock (PR-016), follow with the most active confrontation (PR-008), and close the opening trio with a recognizable protector archetype (PR-027). Three different emotional angles in the first three posts to avoid audience fatigue.

## 03 SCORING & BASELINE BENCHMARKS

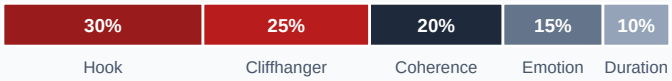
Blind scoring, model-family judge

**Mean composite score by selection strategy**




**Composite score** (weights sum to 1.0):

$$S = 0.30\,s_{hook} + 0.25\,s_{cliff} + 0.20\,s_{coh} + 0.15\,s_{emo} + 0.10\,s_{dur}$$




| STRATEGY | MEAN | vs TOP 15 |
|---|---|---|
| Shortlist (Top 15) | **7.71** | - |
| All Candidates (89) | **6.47** | -1.24 |
| Random 30s | **6.27** | -1.44 |
| Episode Ending | **6.25** | -1.46 |
| Rule-based Heuristic | **6.00** | -1.71 |

## 04 ADVERSARIAL GUARDRAILS & KNOWN FAILURE MODES

Honest engineering

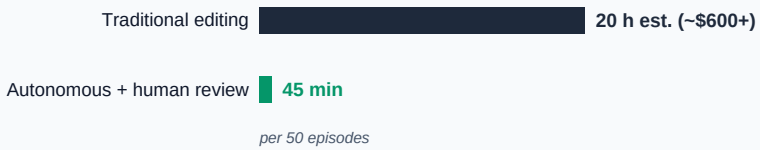
| FAILURE MODE | HOW IT IS HANDLED |
|---|---|
| **Hook Drift Guardrail** | Corrected LLM claims of hooks at **+3.2s** or **+3.9s**. The audit enforces a **hook beat (action or dialogue)** inside **t <= 3.0s**. Beats after 3.0s, like PR-027's shove at +3.2s, do not count. |
| **Premature Resolution vs. True Cliffhangers** | Separates mystery reveals (**PR-016**) from unresolved confrontations (**PR-008**). Flags any cut where the dialogue answers its own suspense loop. |
| **Cold-Start Pronoun Friction** | Flags cuts that open with ambiguous pronouns (*"she"*, *"this"*) before naming the entity, which drives viewer drop-off in the first seconds. |
| **Zero-Hallucination Invariant** | The LLM outputs **element IDs only**. Deterministic Python computes every in/out timestamp, so no timecode is ever generated by a model. |

## 05 GRANULAR PRODUCTION ECONOMICS & UNIT ROI

Compute cost vs. human editing

| SCALE | COMPUTE (USD) | APPROX. INR | NOTE |
|---|---|---|---|
| **Per episode** | **$0.00187** | ~0.16 | <$0.002 / episode |
| **50 episodes** | **$0.0934** | ~7.8 | Total inference spend for this submission |
| **1,000 episodes** | **$1.87** | ~155 | Linear scale-up |
| **Human review** | 20 h to 45 min | ~$600+ est. avoided | >95% labor reduction; ~26x net ROI after review (~6,400x compute-only) |

**Human labor displaced (per 50 episodes, estimated)**




**Unit ROI**

$$\text{ROI}_{compute} = \frac{\$600}{\$0.0934} \approx 6{,}400\times$$

$$\text{ROI}_{net} = \frac{\$600}{\$0.0934 + \$22.50} \approx 26\times$$

Net adds 45 min review at implied $30/h (~$22.50). Human time drops **96%**.

## 06 KUKU FM PLATFORM EXTENSIBILITY

One engine, four product surfaces

**01**
### Audio Drama Trailers
Extends to audio stories for 30s trailers; needs timestamped transcripts and audio-only hook scoring.

**02**
### Ad Creative Testing
Tags cuts by archetype (Revenge vs. Survival) to seed CTR A/B tests on Meta / Reels. Lift untested.

**03**
### Snackable Video Feed
Vertical preview clips deep-linking to full episodes. Needs video assets.

**04**
### Contextual Push Hooks
Cliffhanger lines replace static push copy, spoiler-gated by the audit stage. Context signal TBD.